MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

SEMIOLOGY OF GONADS

Pr. Ioana HRISTOV

SEMIOLOGY

Module: ENDOCRINE SEMIOLOGY

Academic Year : 2026-2027

MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

www.um6ss.ma

Learning Objectives

TESTICULAR DYSFUNCTION SEMIOLOGY

- **ETIOLOGY**
- **SIGNS& SYMPTOMS**
- **DIAGNOSIS**

OVARIAN DYSFUNCTIONS SEMIOLOGY:

-ETIOLOGY

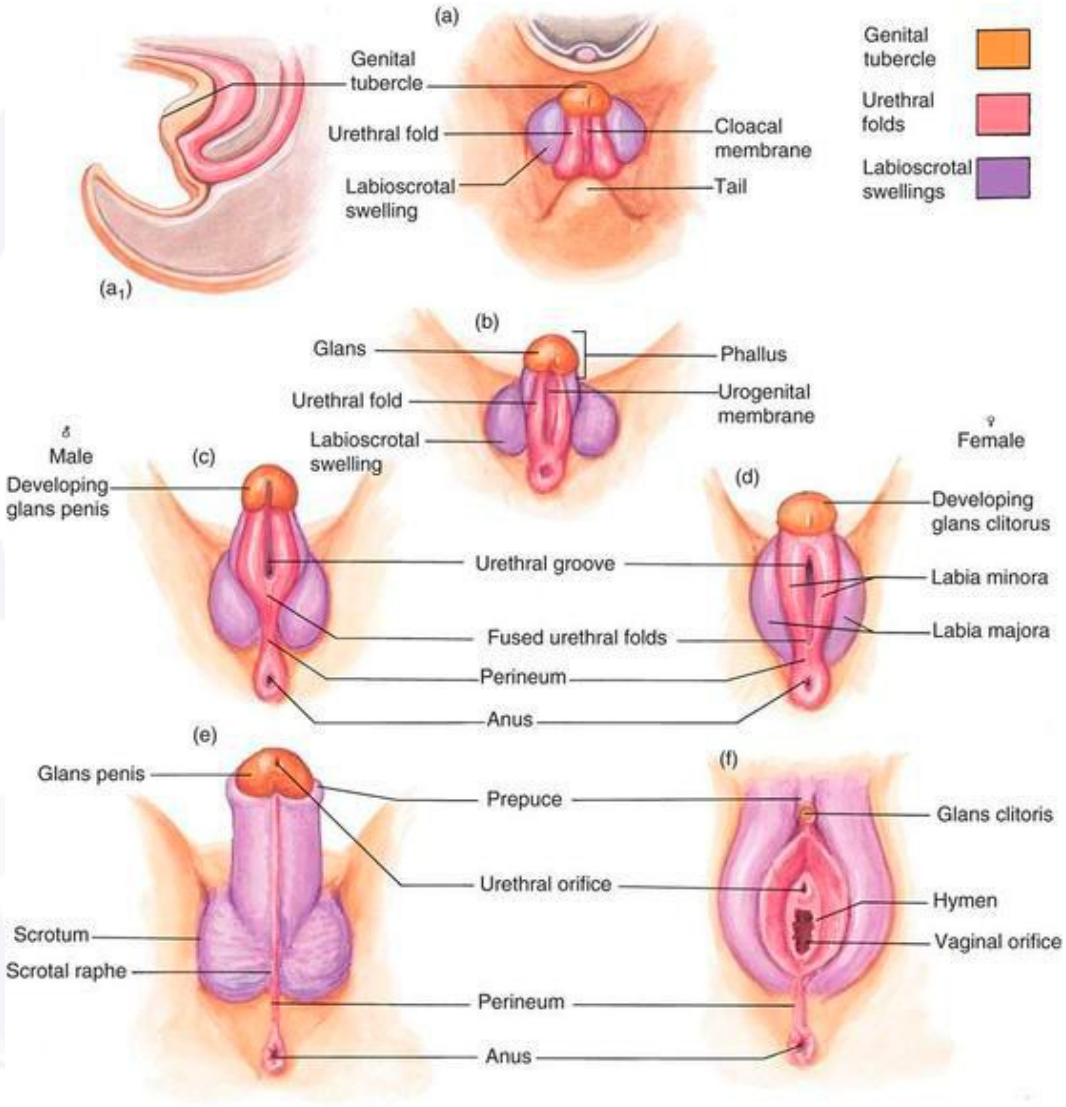
- SIGNS & SYMPTOMS
- DIAGNOSIS

-

Introduction

- The gonads are the glands responsible for the reproductive function in humans. They are under hypothalamic-pituitary control.
- A disorder in the gonad function will have different symptoms depending on the age of onset.
- A delay in diagnosis can affect the prognosis of the development of the reproductive function and normal growth.

Sexual development for Week 12 of gestation



Sex Has Many Levels

1. Genetic Sex – Chromosomes (X & Y in mammals only)
2. Gonadal Sex – Ovaries/Testis
3. Hormonal Sex – Estrogen/Testosterone
4. Somatic Sex – Body anatomy/physiology
5. Psychological Sex - Sexual identity
6. Sexual orientation

Definition



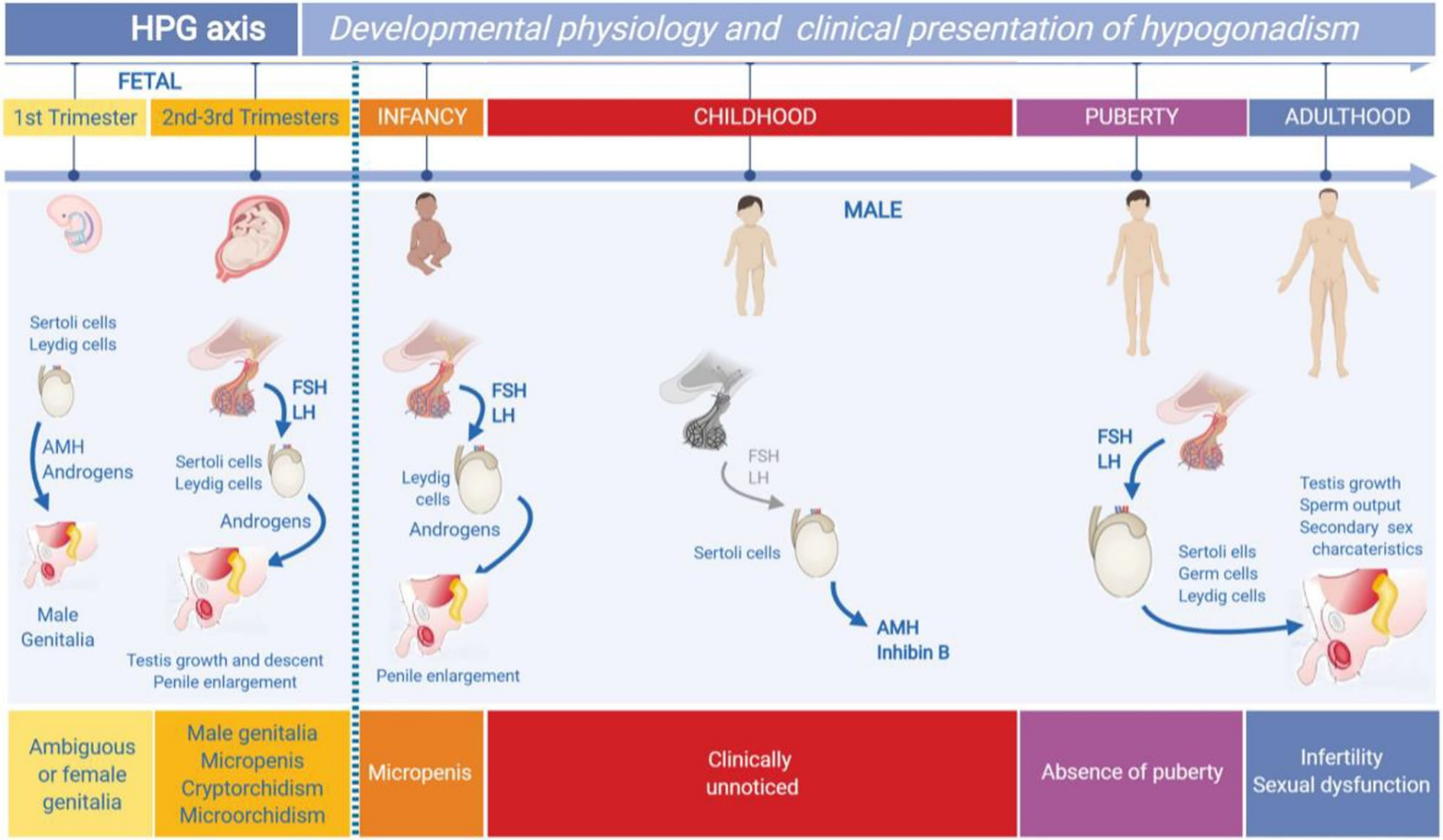
What is hypogonadism ?!

Decreased functional activity of the gonades

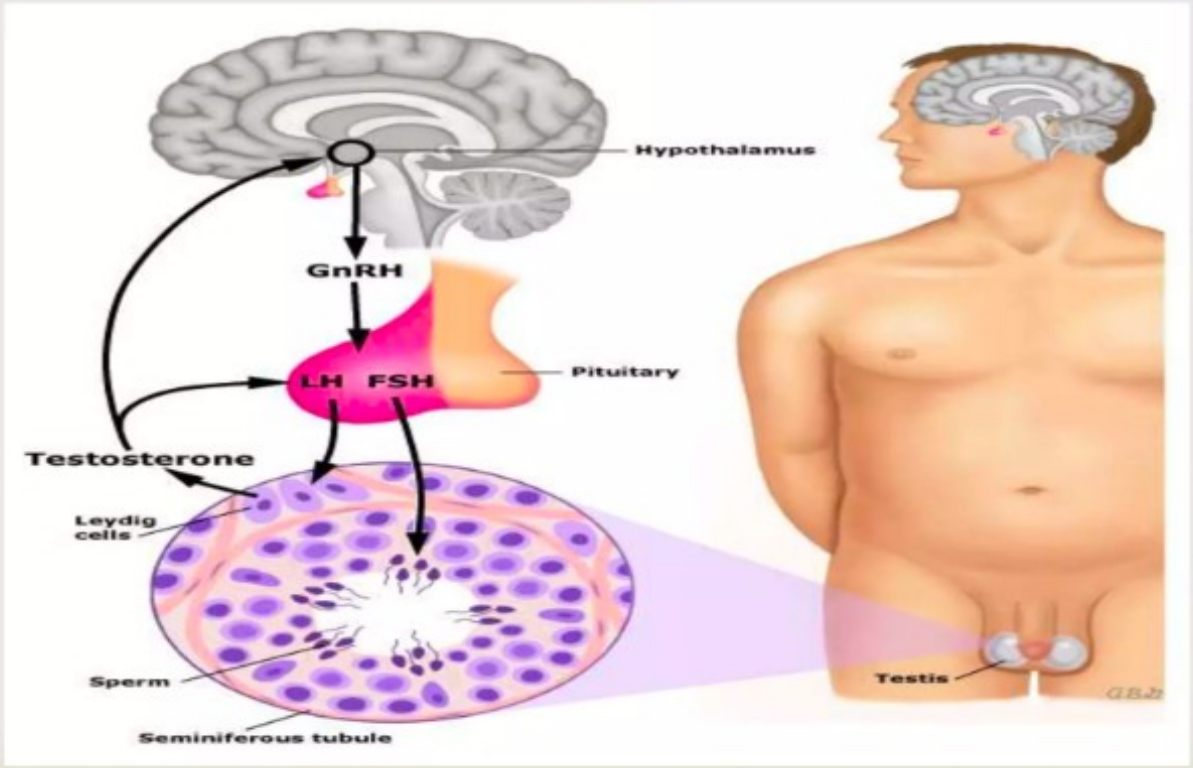
Male hypogonadism ?

Failure of the testes to produce androgen, sperm, or both.

ETIOLOGY : MALE HYPOGONADISM



Hormonal control for male reproduction

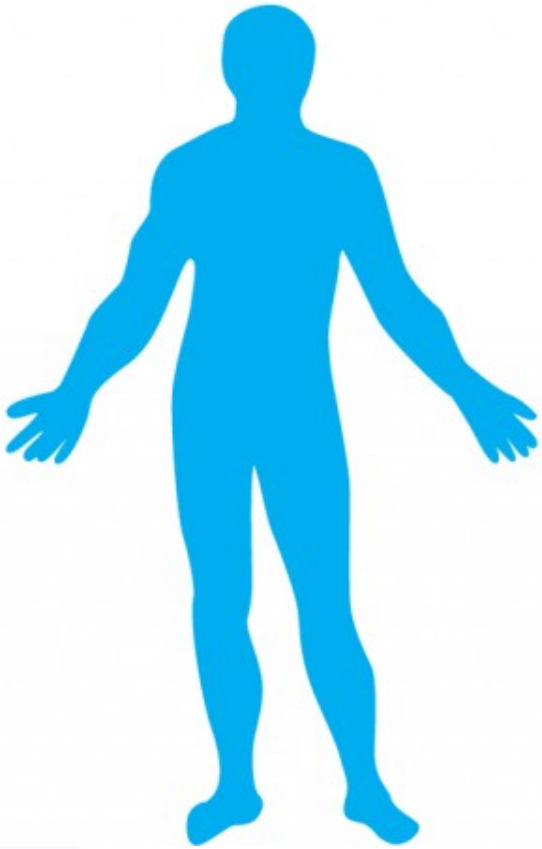


The Effects Of Testosterone

Skin:
Hair Growth
Collagen growth

Sex Organs:
Sperm Production
Erektile Function
Prostate Growth

Muscles:
Muscle Growth
Increased Strength
Increased Endurance



Brain:
Increased Sex Drive
Improved Mood
Confidence
Memory function

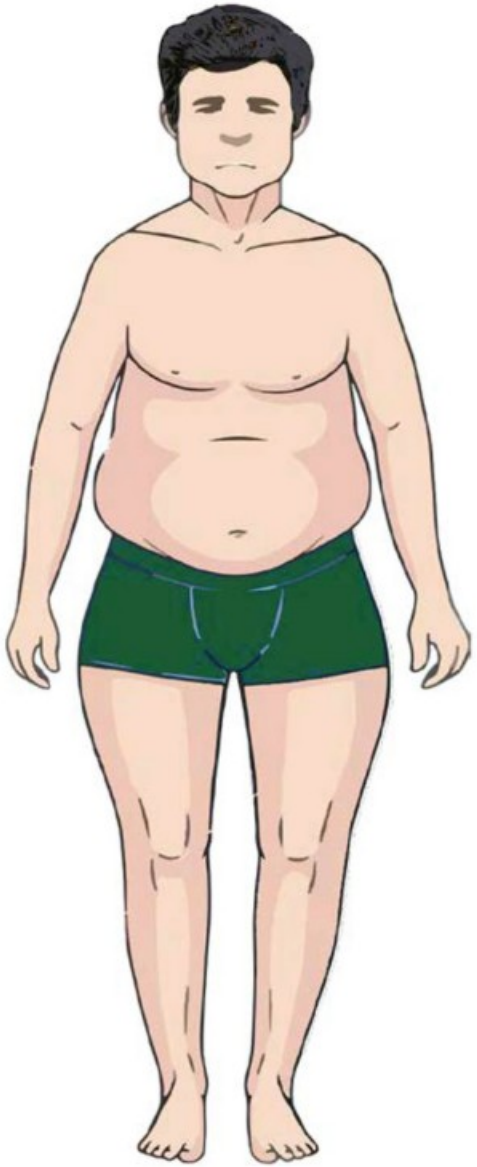
Bone Marrow:
Red Blood Cell
Production

Bones:
Maintenance Of
Bone Mass Density

SEMIOLOGY OF MALE HYPOGONADISM

Symptoms

- Sexual dysfunction
- Low motivation / Vitality
- Poor concentration / Memory
- Hot flushes / sweating
- Infertility



Signs

- Reduced body hair
- Gynecomastia
- Reduced Testicular Volume
- Obesity
  Reduced Muscle Mass
- Anemia
- Reduced Bone Density

ETIOLOGY: MALE HYPOGONADISM CLASSIFICATION

Etiology

Two basic types

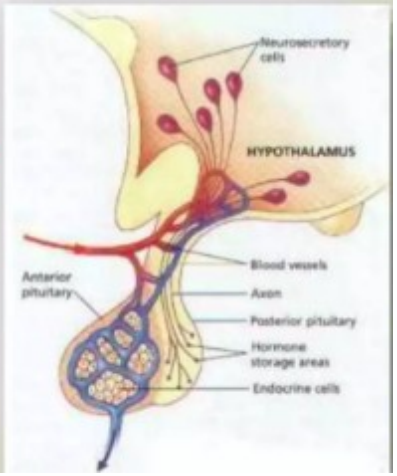
1ry

- ✓ Testicular
- ✓ Receptors



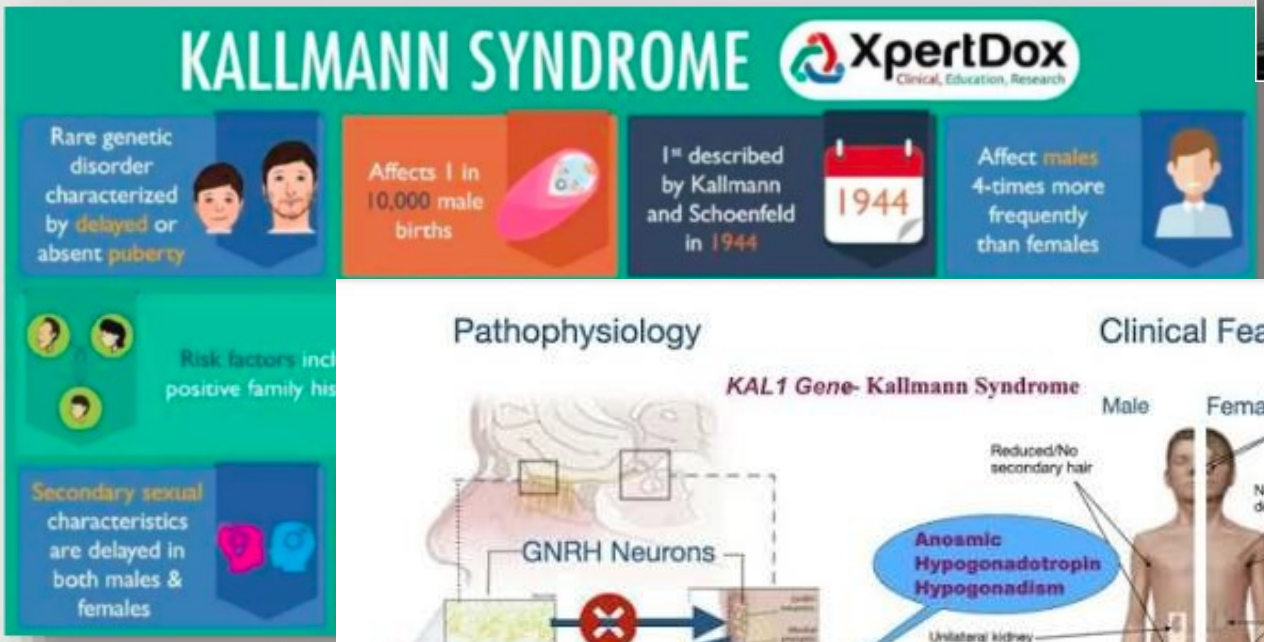
2ry

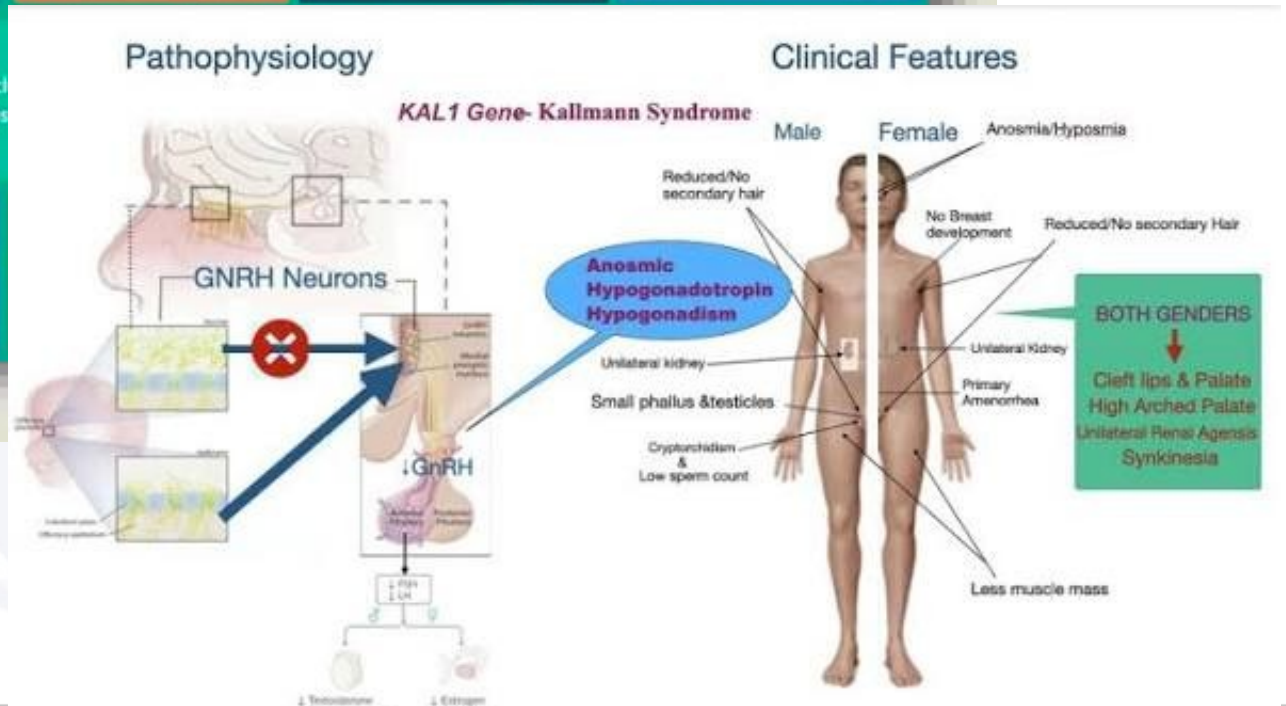
- ✓ Hypothalamus
- ✓ Pituitary



Hypothalamic causes

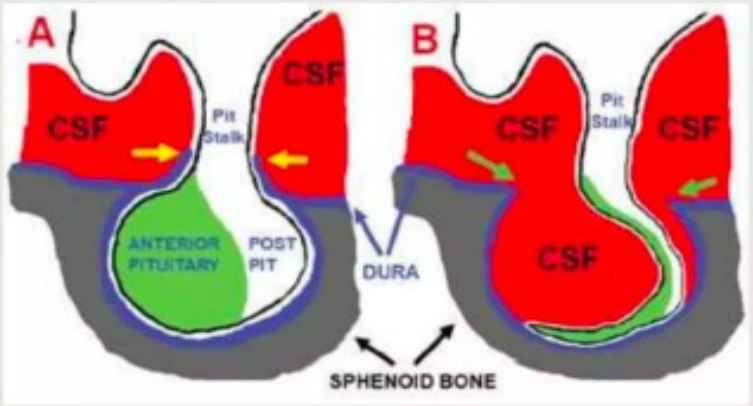






Pituitary causes

Empty sella syndrome

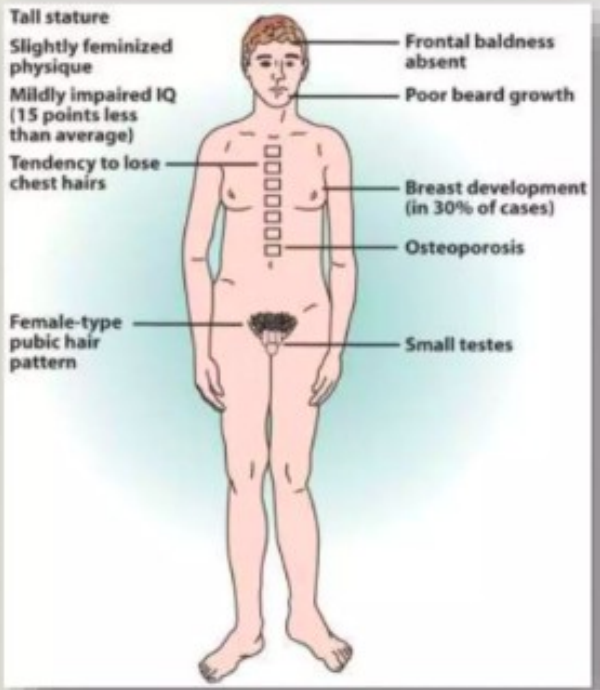


Hyperprolactinemia

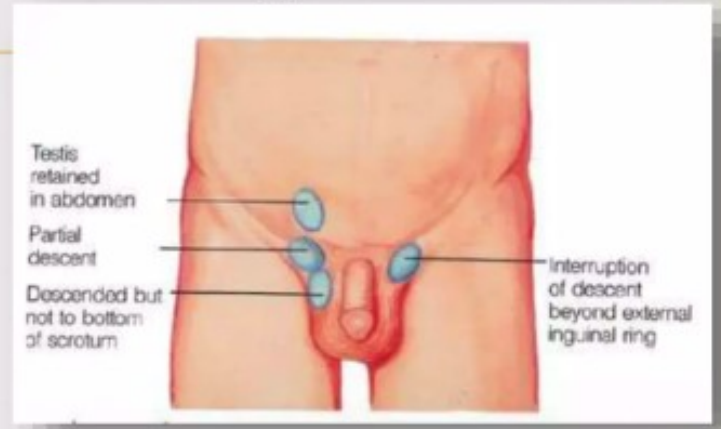


Primary causes

Klinfelter syndrome



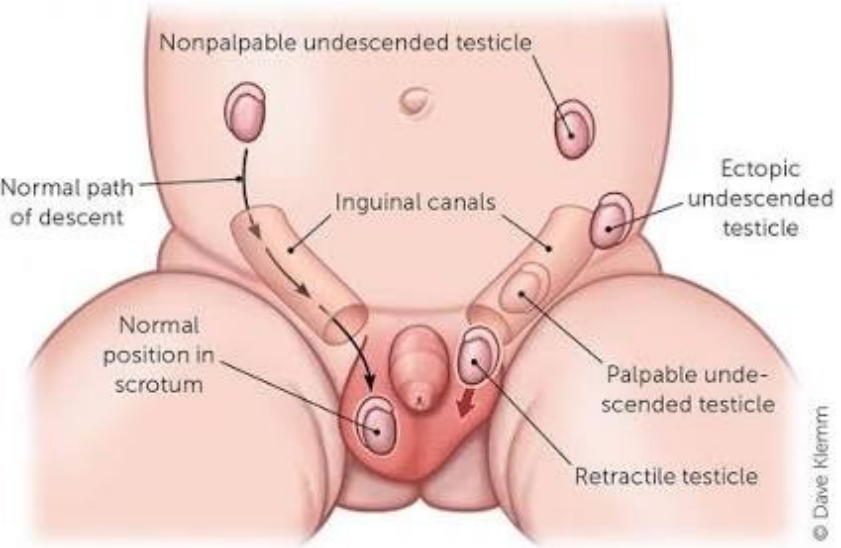
cryptorchidism





Mumps orchitis

Testicular semiological evaluation



Normal position in scrotum: At or below the midpoint of the scrotum without tension.

Retractile testicle: In upper scrotum or lower inguinal canal but can be temporarily repositioned in scrotum without tension.

Palpable undescended testicle: Can be brought into scrotum with persistent tension.

Ectopic undescended testicle: A palpable testicle not located along normal descent.

Nonpalpable undescended testicle: Cannot be clearly palpated.

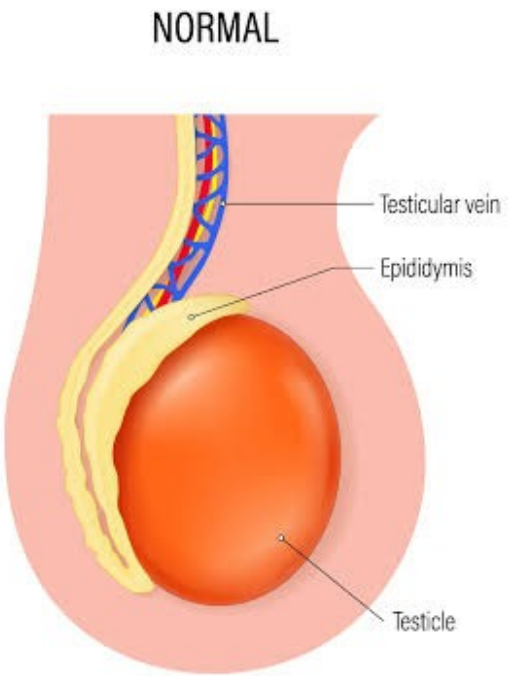


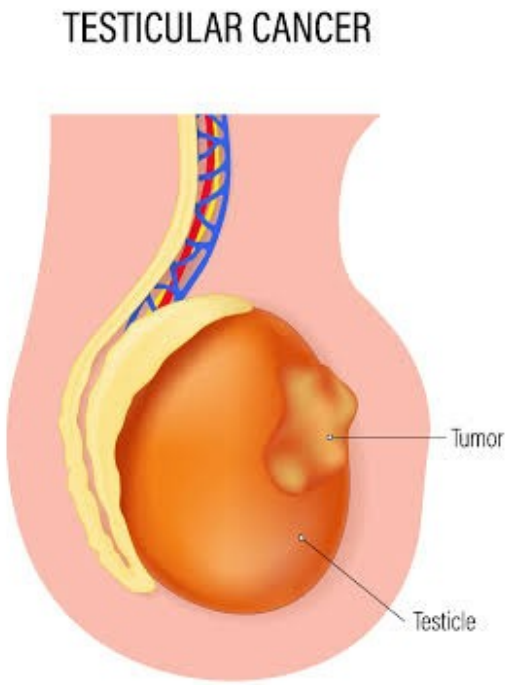
Clinical examination objectives:

1. Right+ left testicule palpable
2. Volume
3. Pain
4. Tumors/ consistency

Testicular semiological evaluation

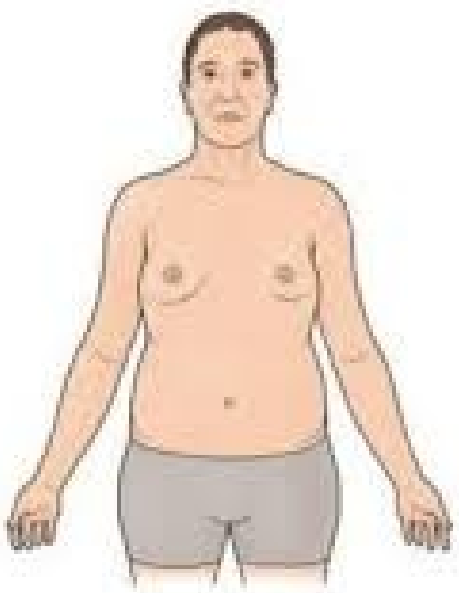


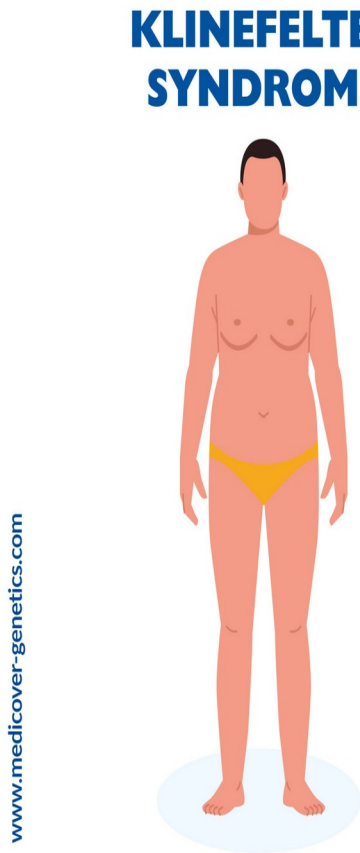


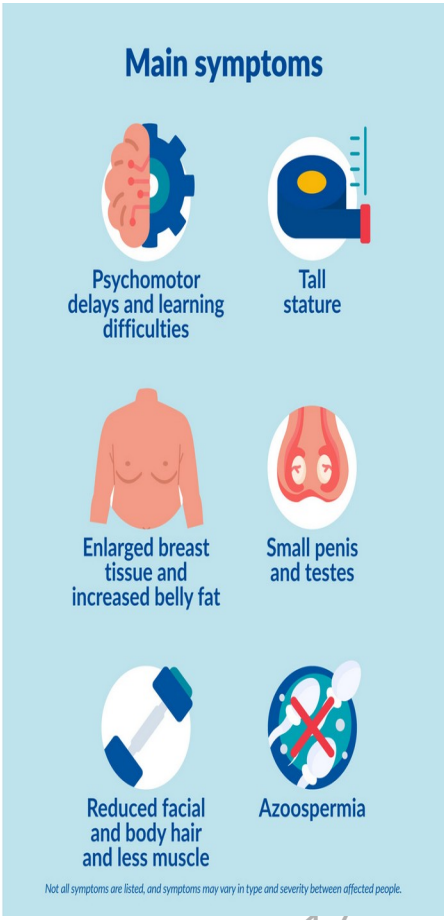


KLINEFELTER SYNDROME

GENETIC KARIOTYPE 47XXY
TESTICULAR FIBROSIS







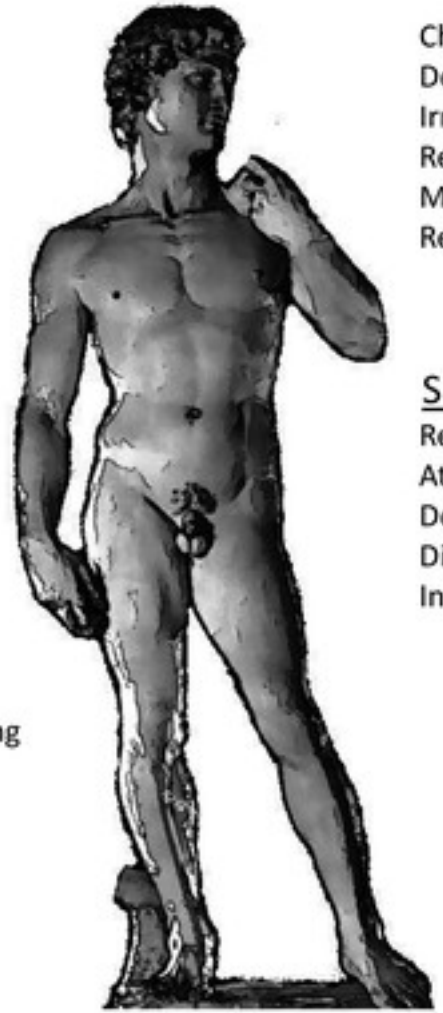
MALE HYPOGONADISM

Signs and Symptoms of Hypogonadism

METABOLIC
Increased risk for Metabolic Syndrome
- Type II diabetes mellitus
- Hypertension
- Lipid derangements
- Truncal obesity

Elevated cardiovascular risk

OTHER
Reduced exercise capacity
Hot flashes
Loss of body hair/reduced shaving
Mild anemia

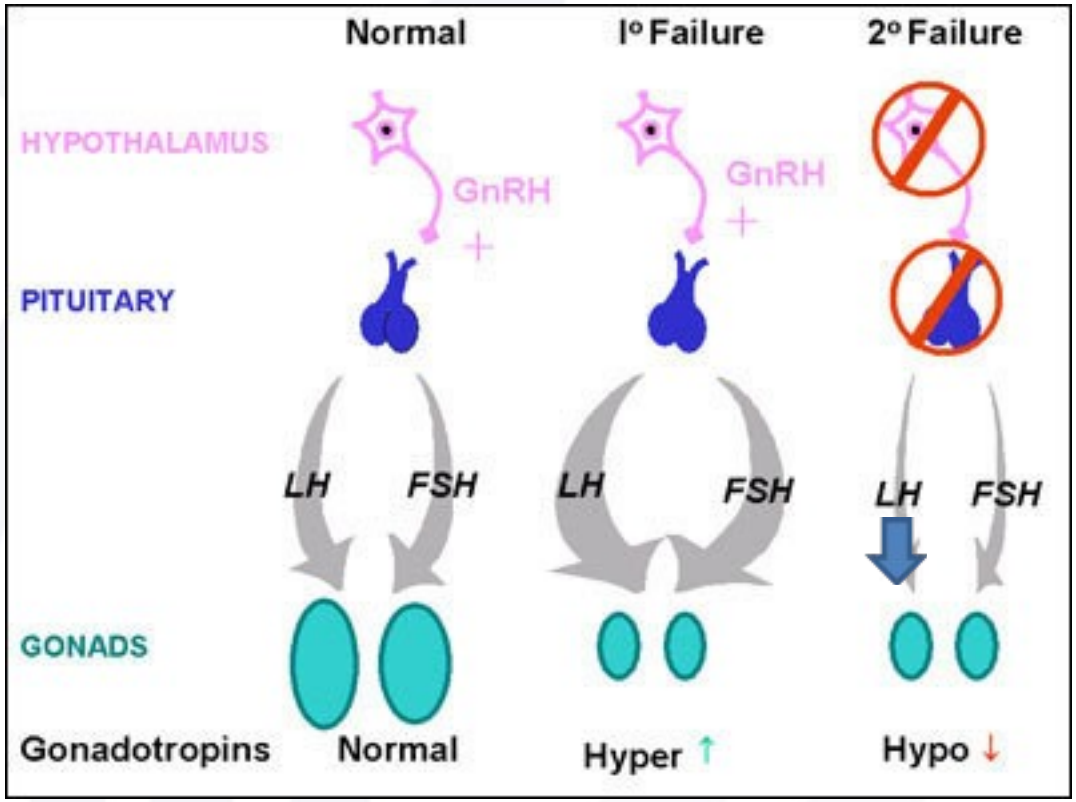


COGNITIVE
Chronic fatigue
Depressed mood
Irritability
Reduced motivation
Memory impairment
Reduced visuospatial performance

SEXUAL AND REPRODUCTIVE
Reduced libido
Atrophy of testicles
Decreased erectile function
Difficulty achieving orgasm
Infertility/Azoospermia

MUSCULOSKELETAL
Reduction in lean muscle mass
Increase in adipose tissue
Elevated risk of osteoporosis
Risk for low trauma fracture

MALE HYPOGONADISM TYPES



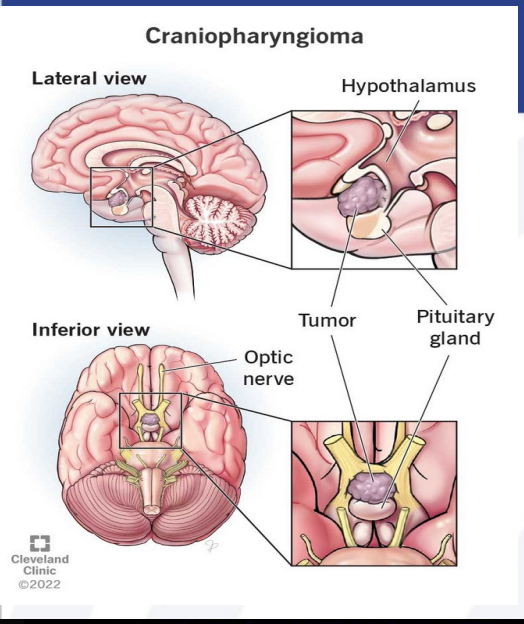
Hypogonadism type	LH, FSH	TESTOSTERON
PRIMARY	↑	↓
SECONDARY	LOW/NORMAL	↓

QUIZ: GINECOMASTIA?



1. 2. 3. 4

ABNORMAL PUBERTAL DEVELOPMENT



PRECOCIOUS PUBERTY (< 8 YEARS)

True precocious puberty (early hypothalamic maturation)	Idiopathic	
	Central nervous system tumors	
Precocious pseudo-puberty	Excess gonadotropin production	Pituitary tumors
		Ectopic (other neoplasms)
	Excess steroid production (usually by "feminilizant" ovarian tumors)	
	Exogenous steroid intake	

DELAYED PUBERTY (> 14 YEARS)

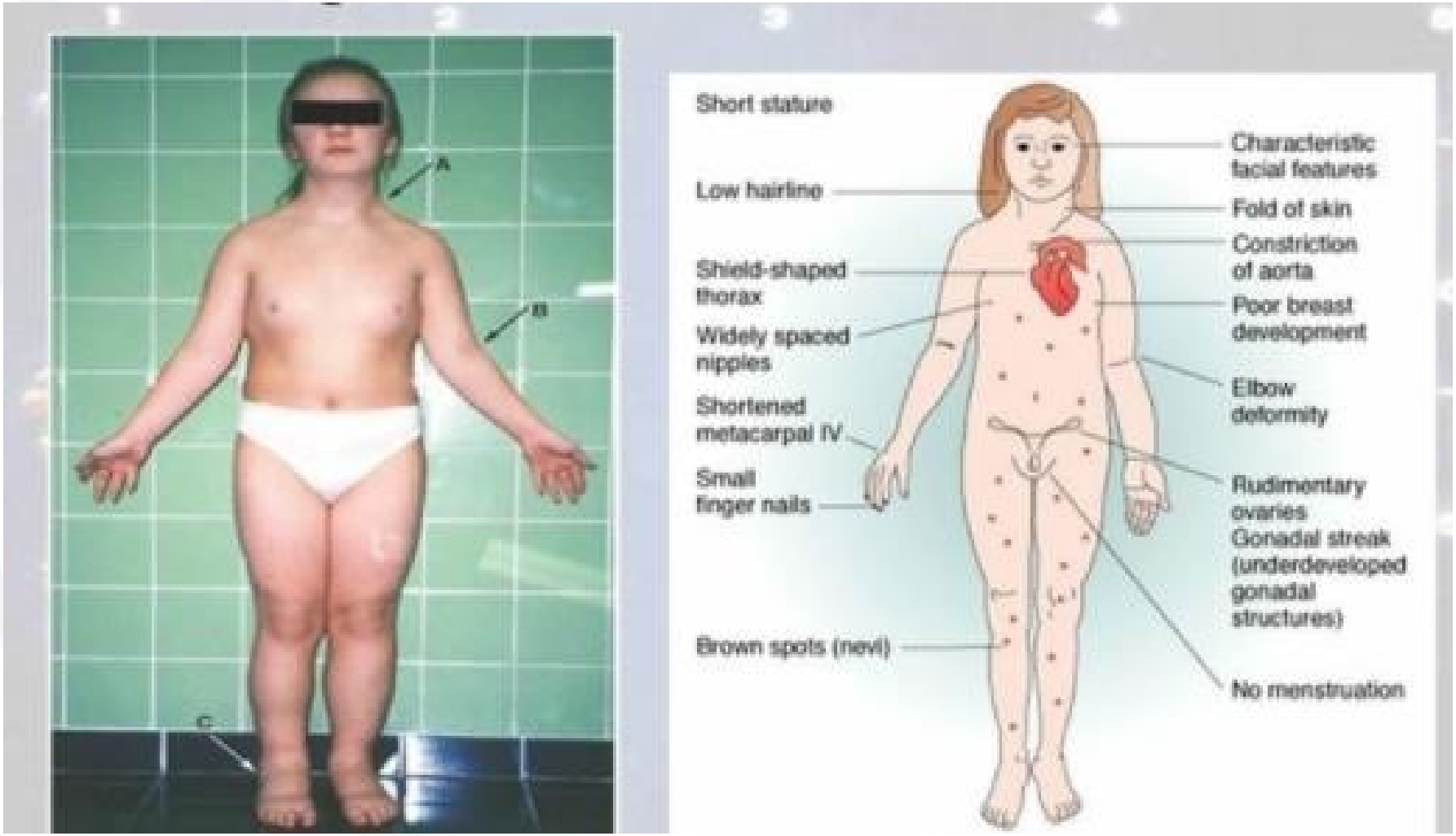
Constitutional (normal late-onset puberty)			
Deficiency of ovarian steroids (hypogonadism)	Hypothalamic/ pituitary deficiency (*hypo*gonadotropic hypogonadism)	Space-occupying lesions	Craniopharyngiomas
			Histiocytosis X
			Pituitary adenomas
		Trauma	
		Functional abnormalities	Kallmann's syndrome
			Panhypopituitarism
			Hyperprolactinemia
			Abnormal body weight
	Primary ovarian deficiency (*hyper*gonadotropic hypogonadism)	Disorders of the sexual differentiation	
		Deficiency of steroid synthesis enzymes	
		Previous radio/chemotherapy	

FEMALE HYPOGONADISM BEFORE PUBERTY: TURNER SYNDROME

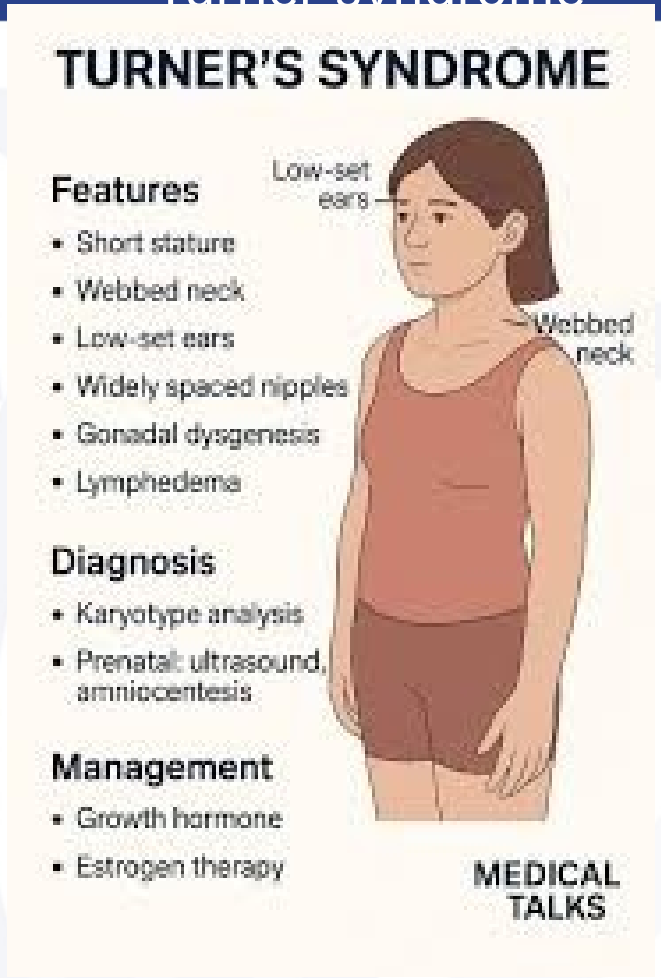
HYPERGONADOTROPIC HIPOGONADISM – GENETIC DISORDERS CARIOTYPE
45X – 1938 – Henry Turner

Estrogens ↓

FSH, LH ↑



Turner syndrome





Low Estrogens- absence of puberty development

Ovarian fibrosis
High FSH, LH

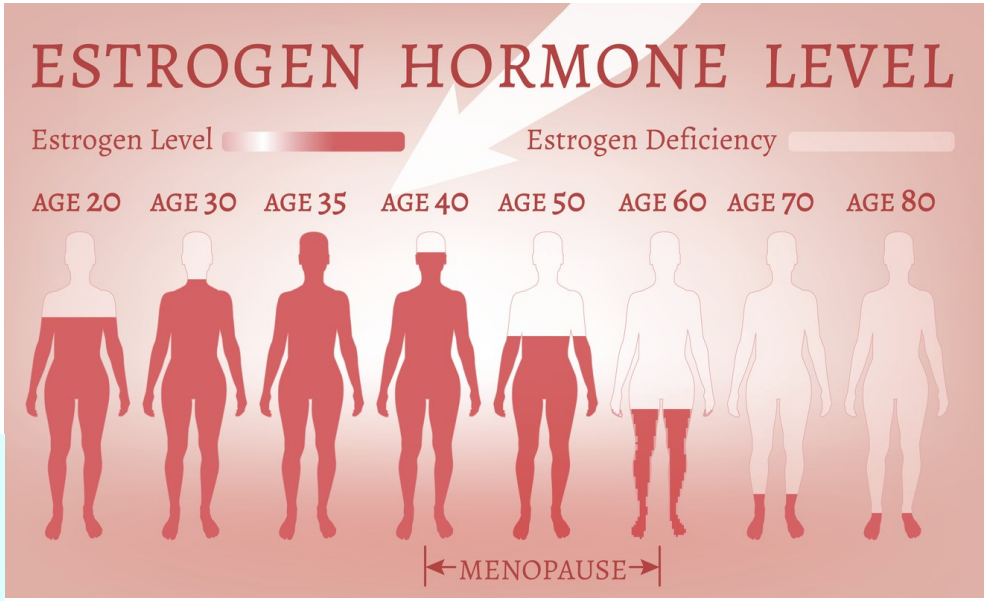
FEMALE HYPOGONADISM

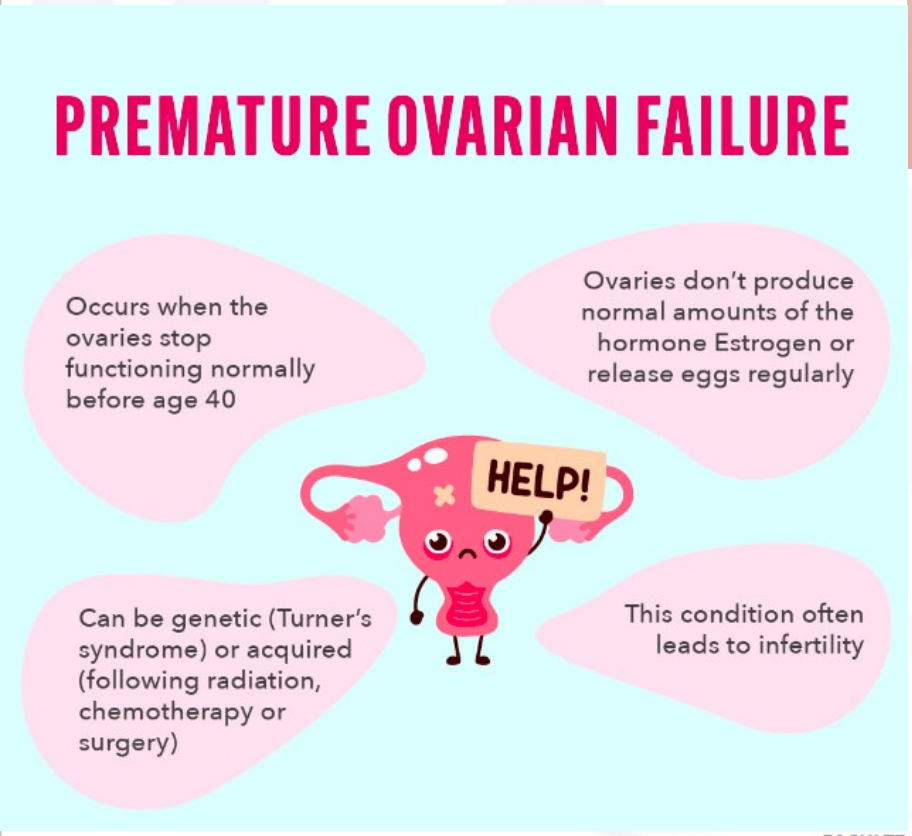
1- clinical findings : after puberty

- **Signs of hypoestrogenism:**
- Incomplete regression of secondary sexual characteristics:
- Loss of vaginal secretions :
- vaginal dryness ± atrophy
- libido
- **Functional urinary discomfort:**
- **-dysuria**
- **-urinary infections**

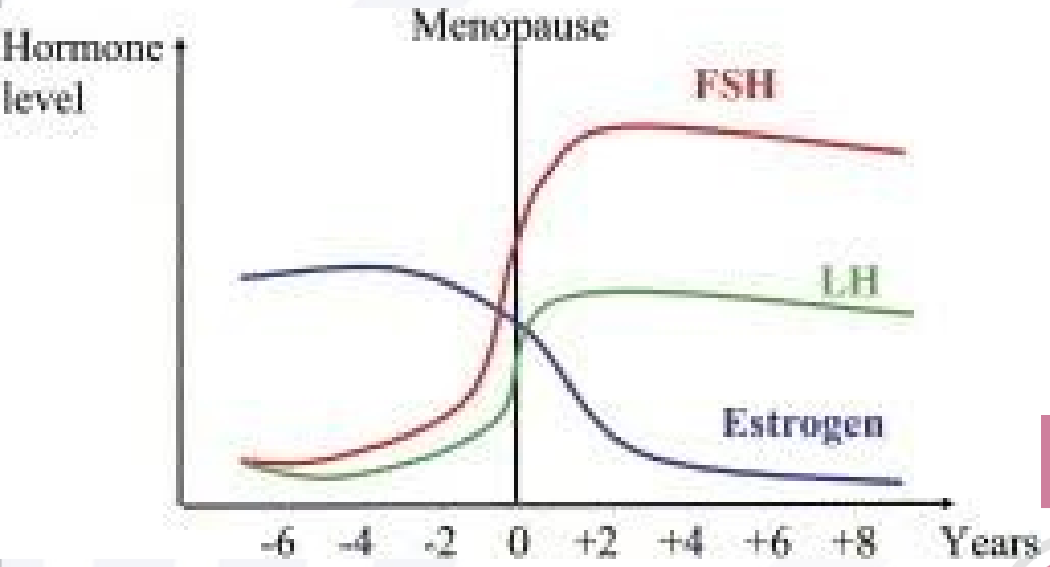
Stress Urinary Incontinence

HYPOGONADISM ETIOLOGY- AFTER PUBERTY





HYPERGONADOTROPIC HYPOGONADISM- OVARIAN INSUFFICIENCY



Main etiology for adult female hypergonadotropic hypogonadism

Low Estrogens

High FSH, LH

Signs of Menopause



Laboratory tests :

Estradiol (E2)

- Reproductive age
  - Estradiol is the primary hormone secreted by the follicle in the ovaries, and it is 12 - 80 times more potent than estrone and estriol
  - Estradiol stimulates development of the female sex organs during puberty, and it causes thickening of the endometrium during the menstrual cycle
  - Normal range: 12.5 - 498 pg/ml depending on the phase of the menstrual cycle
- Menopause
  - Ovarian estradiol production ceases
  - A small amount is produced by the conversion of adrenal steroids in peripheral fat tissue
  - Normal range: 0 - 55 pg/ml depending on the length of menopause

Progesterone

- Reproductive age
  - Progesterone is released by the corpus luteum in the ovaries. It causes the endometrium to become secretory, preparing it for ovum implantation.
  - In pregnancy, it is produced by the placenta during the second and third trimesters
  - Normal range: 0.2 - 27 ng/ml depending on the phase of the menstrual cycle
- Menopause
  - Ovarian progesterone production stops
  - Normal range: 0.1 - 0.8 ng/ml

Testosterone

- Reproductive age
  - Testosterone is produced by the ovaries in small amounts. In the ovaries, aromatase converts testosterone to estradiol.
  - Normal range: 8 - 48 ng/dl
- Menopause
  - Ovaries continue to produce testosterone
  - Normal range: 3 - 41 ng/dl

LABORATORY TESTS

Follicle-stimulating hormone (FSH)

- Reproductive age
  - FSH is released from the anterior pituitary in response to GnRH. FSH stimulates follicles in the ovary to mature.
  - Normal range: 3.5 - 21.5 mIU/ml depending on the phase of the menstrual cycle
- Menopause
  - Estrogen loss reduces negative feedback in the pituitary and hypothalamus, and FSH levels rise
  - Normal range: 25.8 - 135 mIU/ml

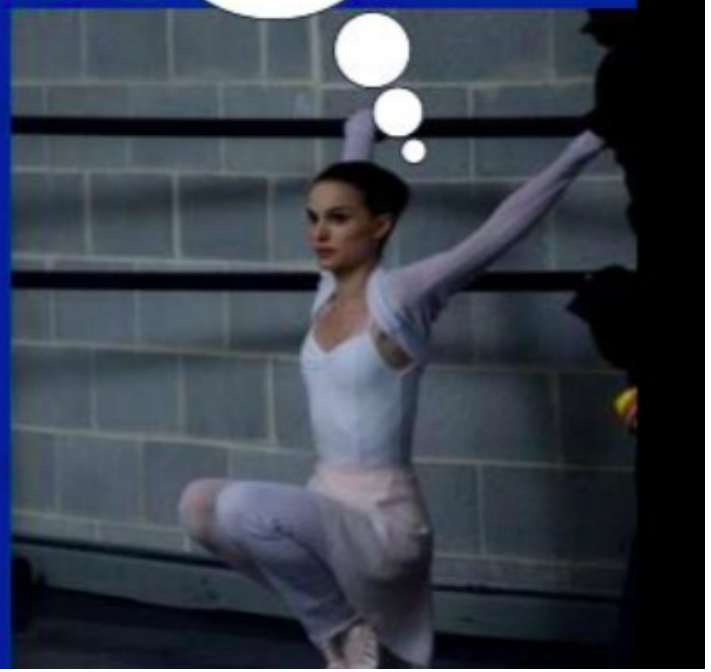
Luteinizing hormone (LH)

- Reproductive age
  - LH is released from the anterior pituitary in response to GnRH
  - The LH surge in the middle of the menstrual cycle stimulates ovulation
  - LH maintains the corpus luteum and stimulates steroid production
  - Normal range: 2.4 - 95.6 mIU/ml depending on the phase of the menstrual cycle
- Menopause
  - Estrogen loss reduces negative feedback in the pituitary and hypothalamus, and LH levels rise
  - Normal range: 7.7 - 58.5 mIU/ml

CLINICAL SCENARIO

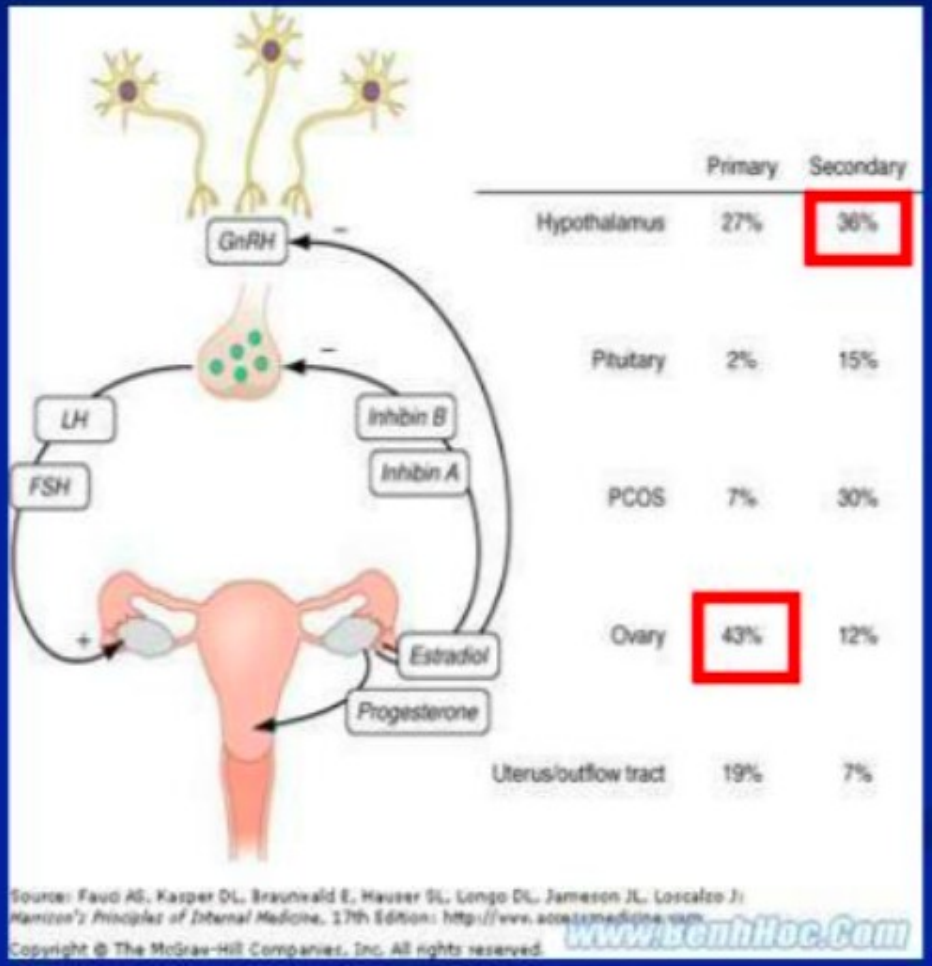
- 18 yo G0P0 woman referred to your clinic:
  - **"I haven't had my period for 6 months"**
  - Menarche at age 12, normal 2° sex characteristics, no sexual activity
  - Daily training for National Ballet School audition and has lost 5lbs (BMI 19)
  - Home-school, Mother is strict but supportive





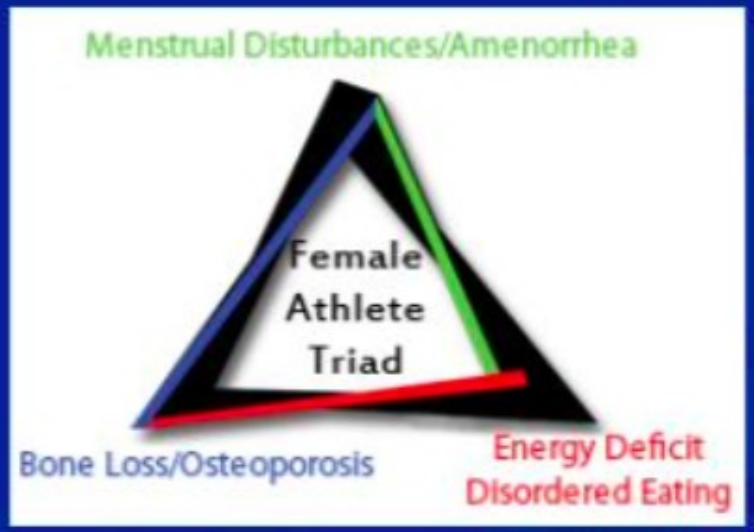
H-P-O AXIS & MENSTRUATION

- Hypothalamus
- Pituitary
- Ovaries
- Uterus and outflow tract



HYPOTHALAMIC AMENORRHEA

- Secondary amenorrhea due to suppression of H-P-O axis via GnRH pulsatility
- No anatomic or organic disease = Diagnosis of Exclusion
- STRESS
  - Energy deficit
  - Wt loss, eating disorder
  - Excessive exercise
  - Psychological
- Genetic?



HYPOGONADOTROPIC HYPOGONADISM

- **Functional Hypothalamic Amenorrhea**
  - **Anorexia or bulimia nervosa**
  - **Excessive exercise**
  - **Excessive weight loss or malnutrition**
- Hypothalamic or pituitary destruction
- Central nervous system tumor
- Constitutional delay of growth and puberty*
- Chronic illness
  - Liver disease, Renal insufficiency, Diabetes, Immunodeficiency, Inflammatory bowel disease, Thyroid disease, Severe depression or psychosocial stressors
- Cranial radiation
- Congenital GnRH deficiency*, Kallmann syndrome*
- Sheehan's syndrome

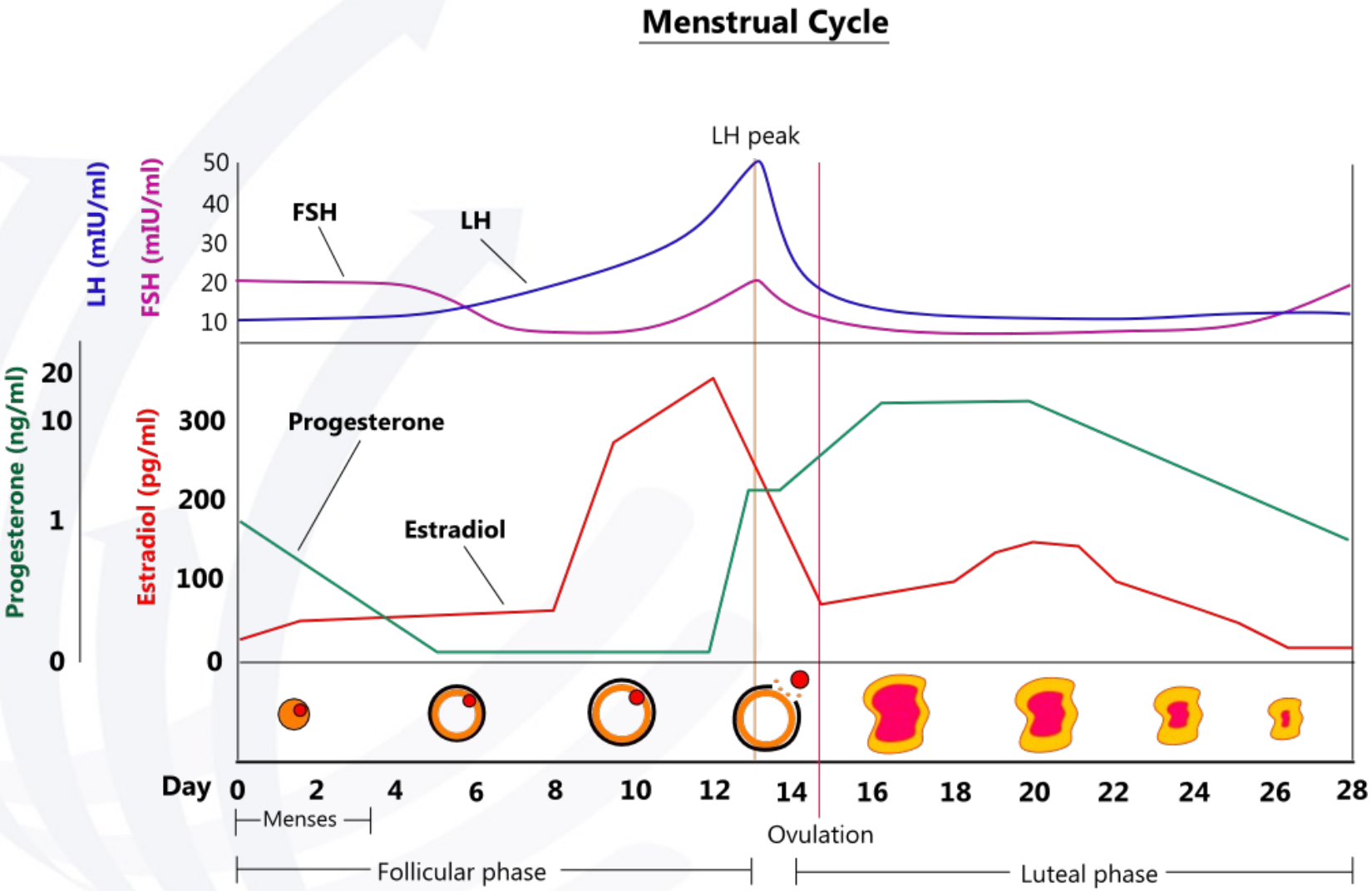
*causes of primary amenorrhea only

INVESTIGATIONS

1. Rule out pregnancy – βhCG
2. Hypercortisol – Cortisol AM, ACTH
3. Hypothyroid – TSH, FT3, FT4
4. Prolactinoma – Prl, MRI
5. Ovarian insufficiency – FSH, LH
6. Hyperandrogenism – Free testosterone, DHEAS
7. Chronic systemic illness – CBC, Ferritin, ACE, FBG, HbA1C, Karyotype, BMD, 25-OH Vit D, LFTs, albumin, lipid profile

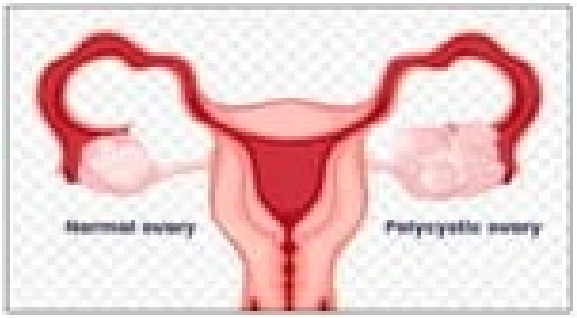
↓Estradiol, ↓/low-normal LH and FSH

MENSTRUAL CYCLE



Polycystic ovary syndrome (PCOS)

- Clinically defined as hyperandrogenism with chronic anovulation without underlying disease of adrenal or pituitary glands.
- Signs and symptoms :
  - Amenorrhea
  - Heavy vaginal bleeding
  - Hirsutism
  - Acne
  - Infertility
  - Obesity→ OVERWEIGHT
- Insulin resistance, dyslipidemia, glucose intolerance and type 2 diabetes are strongly associated



TWO of three criteria for PCOS

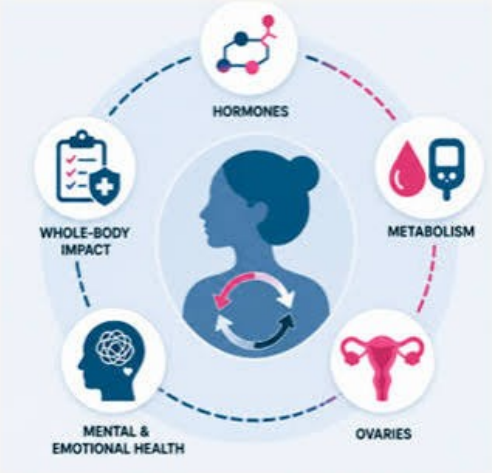
- Menstrual irregularity
- Clinical or biochemical androgen level increased
- Multiple cyst in ovaries.

REACH FERTILITY

A NEW NAME- A DEEPER UNDERSTANDING

PCOS → PMOS

What Changed?



PCOS – Clinical Features

- **Hyperandrogenism**
  - Hirsutism
    - Modified Ferriman Gallwey Score
  - Acne
  - Androgenic alopecia
- **Menstrual Irregularity**
  - Oligomenorrhea (70-75%)
  - Amenorrhea (20%)
  - Infertility (30-70%)





ALWAYS LOOK FOR GONADS DYSFUNCTION IN CASE OF INFERTILITY

CAUSES OF INFERTILITY



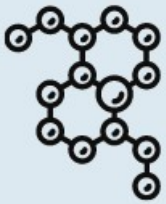
Ovulation Disorders



Age



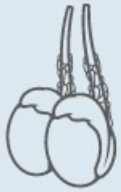
Uterine fibroids
& Endometrial polyps



Hormone Imbalances



Infections



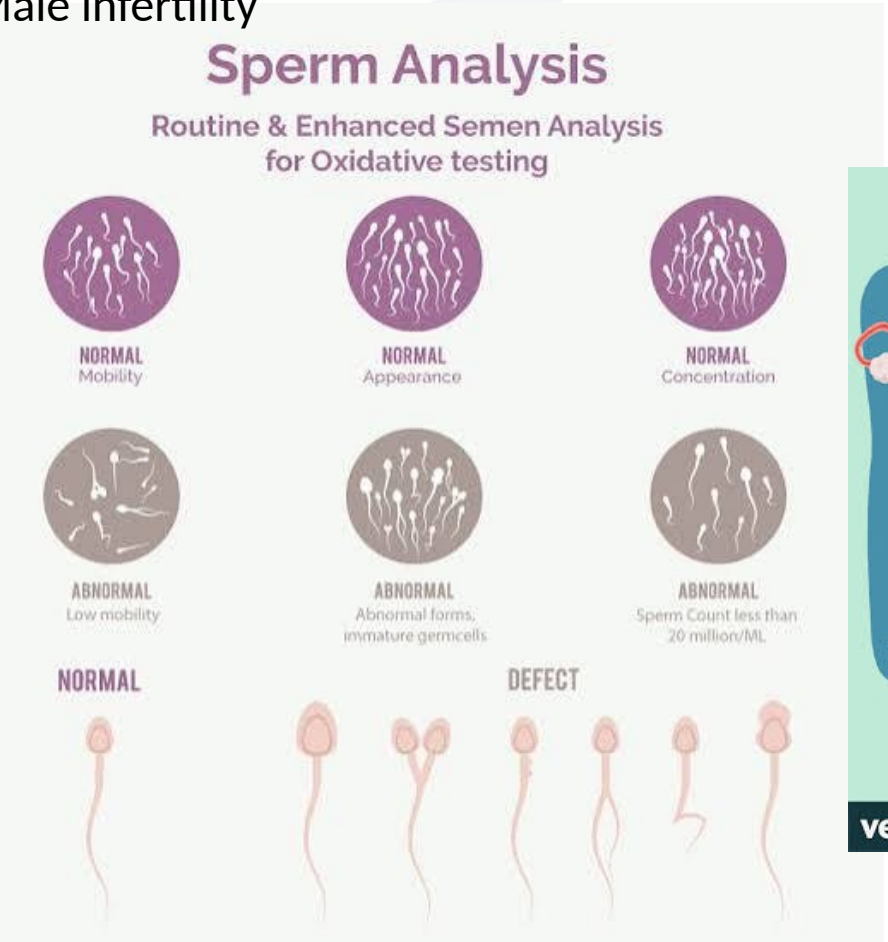
Varicocele



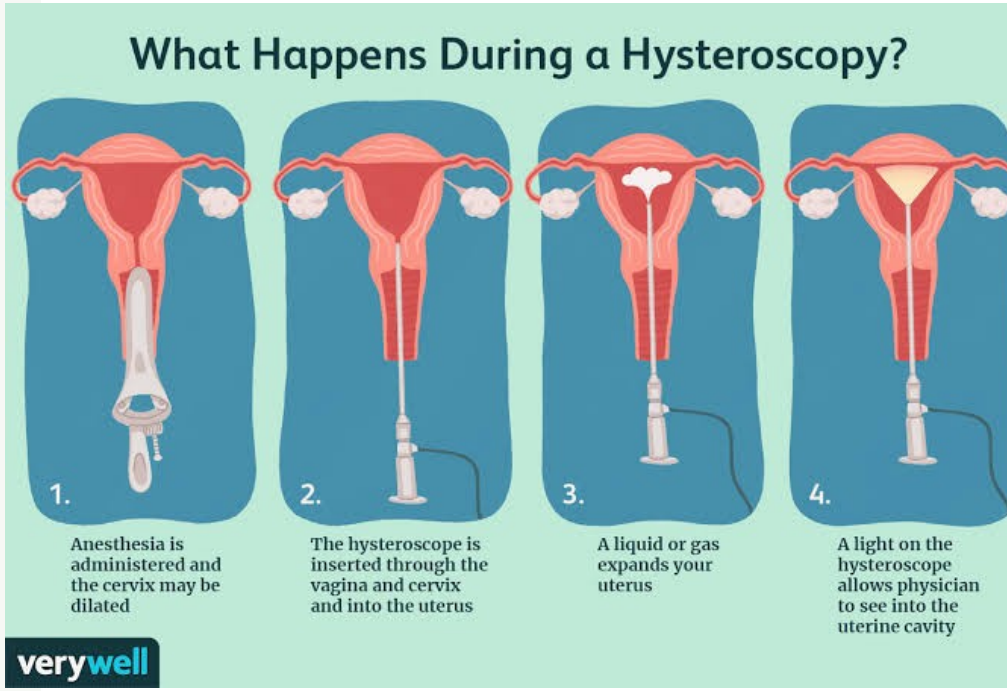


Main infertility investigation

Male infertility



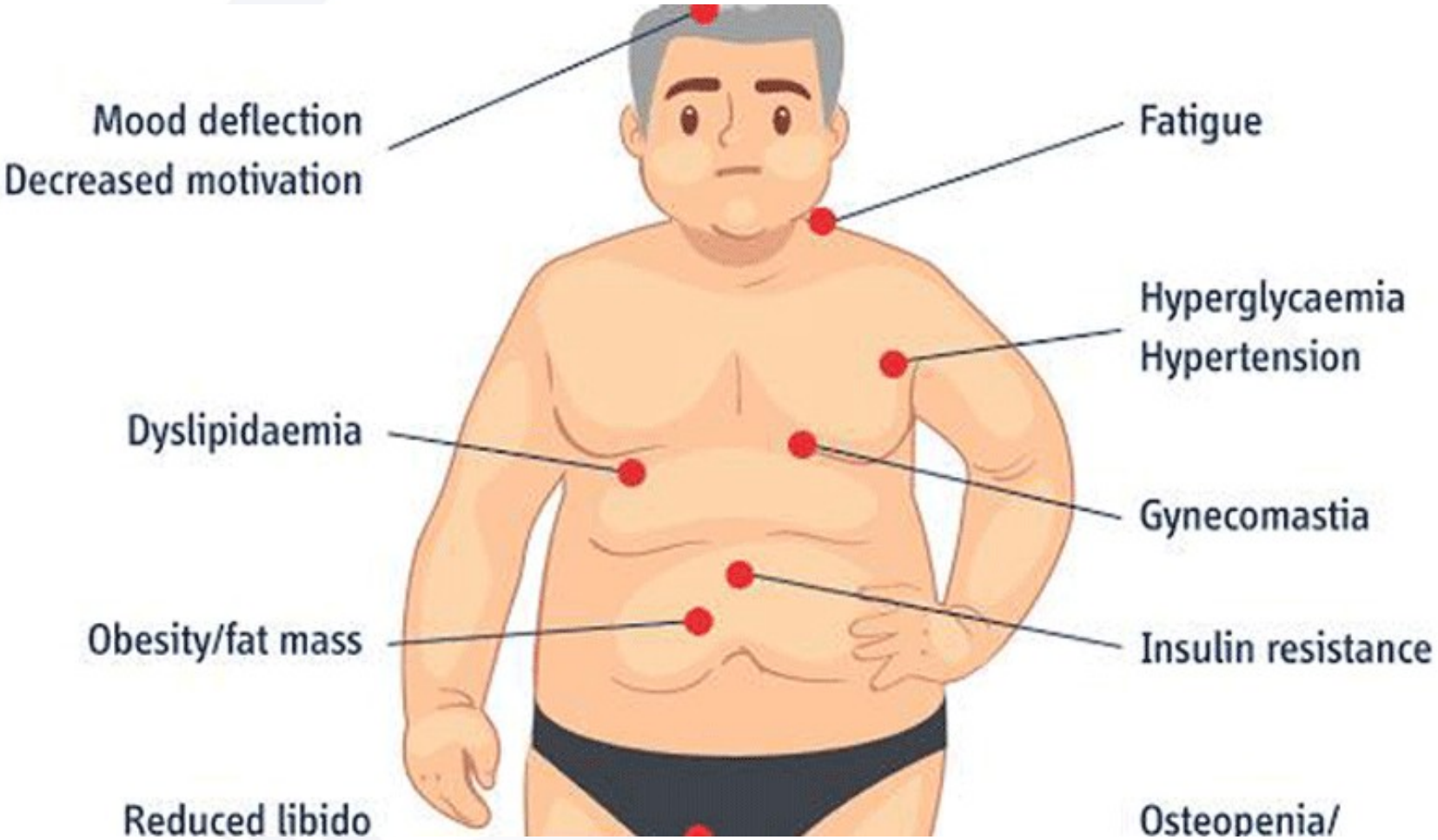
Female infertility



Hormonal evaluation:
FSH, LH, Prolactin,
Estradiol/ progesteron balance
Testosteron/ Inhibin B

Take home message

Male hipogonadism



Take home message

Most frequent female Gonadal dysfunction

